



Interaction designer with an engineering background, creating playful, user-centered experiences across digital, physical, and emerging technologies.

Selected Projects

KONE Concierge

Spring 2026

AI Experience Design

- Designed an AI concierge for hotel elevators, translating research on shared-space behavior into principles for **when and how an AI should engage**.
- Designed interaction flows balancing **relevance, privacy, and user autonomy**, informed by interviews, storyboard evaluations, and desktop research.

Hesper, the Artsy Ghost: MR Museum Guide

Spring 2026

MR Interaction Design

- Designed an mixed reality museum guide that lets visitors explore artwork interpretation at their own pace.
- Conducted observational research and iterative user testing to develop an **autonomy-preserving interactive system**.

Frictionless: Plasma brake test rig

Fall 2025 - Spring 2026

Product Design

- Prototyped a monitoring control system for a plasma brake deployment test rig, **translating ambiguous requirements into an operable system**.
- Built the Raspberry Pi control architecture and a lightweight real-time GUI.
- Designed an interactive exhibition game to make the technology tangible to non-technical audiences.

Catcha: Toy dispenser for house cats

Winter 2024

Animal-centered Design

- Designed and prototyped a self-operated toy dispenser around a cat's physical abilities and behavioral preferences, iterated through cardboard mechanisms and user testing.

More details and process: michelle-y-chang.com

Research

Aalto University | Ambient Intelligence Group

January - July 2026

Research Assistant — *Multi-user RFID-based Gesture Recognition with Federated Learning*

- Developed and evaluated a federated learning algorithm for privacy-preserving gesture recognition, demonstrating performance between centralized and local-only approaches.
- Investigated challenges in RFID-based gesture recognition for multi-user settings.
- Designed and conducted a multi-user gesture data collection study, building a dataset for model development and evaluation.

Skills

Design

Product Design, Interaction Design, UX/UI Design, User Research, Usability Testing, User Flows, Design Systems, Rapid Prototyping, Software Prototyping, Hardware Prototyping, Physical Computing, Figma

Tech

Python, Java, C++, C, SQL, TypeScript, Unity, C#, HTML/CSS, Arduino, Mjoco

DevOps / Tooling

Git, Jira, Docker, AWS, Jenkins, CI/CD

Education

KTH Royal Institute of Technology

2026 - 2027

M.Sc. in Human-Computer Interaction Design

Focus: physical interaction · multimodal interfaces · social robotics

Aalto University

2025 - 2026

M.Sc. in Human-Computer Interaction Design

Focus: product design · experience design · emerging interfaces · rapid prototyping

Carnegie Mellon University

2017 - 2021

B.Sc. in Electrical and Computer Engineering, minor in Game Design

Graduated with University Honors

Focus: signal processing · embedded systems · computer architecture · computer graphics

Capstone project: FPGA-based music synthesizer

Professional Experience

Intuit

2021 - 2024

Software Engineer

- Developed **scalable backend services and APIs** for QuickBooks Online, contributing to production systems supporting millions of users.
- Built an AI-driven pipeline that parsed requirements into executable interface-level simulations, enabling end-to-end validation with penny-level accuracy.
- Monitored production traffic and resolved inefficiencies to improve system caching performance by 40%.

Intuit

2020

Software Engineer Intern

- Worked alongside an engineering team to develop an internal productivity tool — a resilient configuration management library supporting dynamic updates with local fallback.